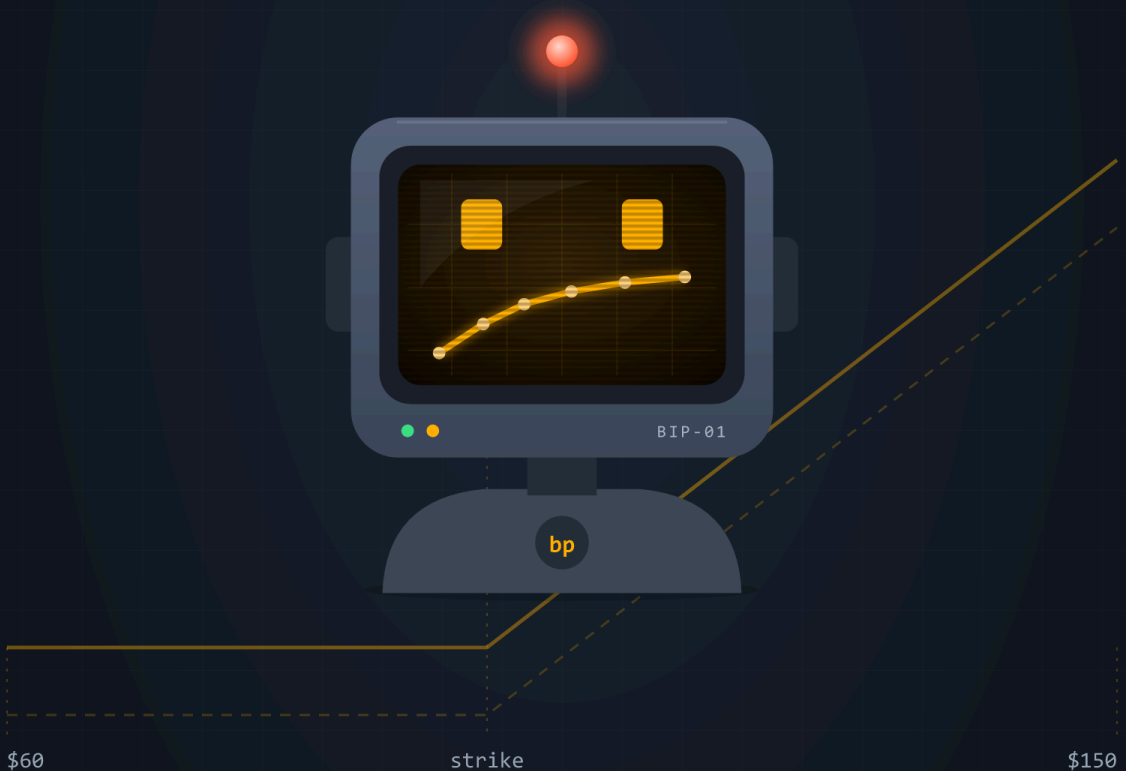


OPTIONS FROM ZERO • PART 2

Options from Zero Part 2

Hedging, the Greeks and your own
markets, with Python and QuantLib



Rates with Bip

OPTIONS FROM ZERO • PART 2 OF 3

Options from Zero

Part 2: Hedging, the Greeks and your own markets, with Python and QuantLib

Rates with Bip

Options from Zero. Part 2: Hedging, the Greeks and your own markets, with Python and QuantLib.

© 2026 Rates with Bip. This book is licensed under the Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International licence (CC BY-NC-ND 4.0): you may copy and share it unchanged, for non-commercial purposes, crediting Rates with Bip. Licence: creativecommons.org/licenses/by-nc-nd/4.0. The companion notebooks are free under the MIT licence at github.com/RatesWithBip/curriculum.

The share in this book, its option prices, its volatility and the interest rate are illustrative, made up for teaching, and every simulated path is simulated. They are not market data. The AUD/USD, SOFR, AONIA and BBSW quotes in Chapters 10 and 11 are the illustrative market of the Series 2 book. The AUD/USD exchange rates in Chapter 9 are the noon buying rates in New York from the Federal Reserve Board's H.10 release. Source: Board of Governors of the Federal Reserve System.

AUD/USD option conventions are checked against Reiswich and Wystup (2009), and the history of negative policy rates against the European Central Bank's and the Bank of Japan's own publications; sources are listed at the end of each chapter. Options are risky: a seller can lose far more than the premium received.

Every number in the text, tables and figures is computed by the companion QuantLib Python notebooks (QuantLib 1.43), one per chapter, which reproduce each result.

This book is for education only. It is not financial, investment, legal or tax advice.

Contents

How to use this book	5
1. Random Walks and Brownian Motion	7
2. Slopes, Curvature and Taylor Series	10
3. Itô's Lemma, Without the Pain	13
4. The Black-Scholes Equation from Hedging	16
5. Delta, Gamma, Theta	19
6. Vega, Rho, and Explaining a Day's P&L	22
7. Monte Carlo Pricing	25
8. Hedging in Practice: Implied vs Realised Volatility	28
9. Implied Volatility: The Price Is Quoted as a Vol	31
10. FX Options: Garman-Kohlhagen	34
11. Rates Options: Black-76 and Bachelier	37
Checkpoint and glossary	41

How to use this book

This book follows the eleven video episodes of Part 2 of *Options from Zero*, one chapter per episode, and goes a little deeper in each. It picks up where Part 1 ended: you know what calls and puts are, how a binomial tree prices them, and how to read each term of the Black-Scholes formula. Part 2 gets the same formula a second way, from hedging, and then uses it the way a trader does.

- **Chapters 1–3** build the maths of randomness: random walks and Brownian motion, a pit stop on slopes and curvature, and Itô's lemma.
- **Chapter 4** derives the Black-Scholes equation from a hedged portfolio and solves it on a grid.
- **Chapters 5–6** are the Greeks: delta, gamma, theta, vega and rho, and explaining a day's profit and loss with them.
- **Chapter 7** prices by simulation: Monte Carlo and its error bars.
- **Chapter 8** hedges an option for a year and shows that a hedged option is a bet on volatility.
- **Chapter 9** runs the formula backwards: implied volatility.
- **Chapters 10–11** carry everything into the reader's own markets: options on the Australian dollar (Garman-Kohlhagen) and caplets on BBSW (Black-76 and Bachelier).

Each chapter ends with the chapter's one idea to remember, which the videos call *Bip's rule*, and a question to try before the next chapter. The answer opens the next chapter. The appendix has the Part 2 checkpoint with worked answers, and a glossary.

Numbers you can reproduce

Every number in the text, tables and figures is computed by a companion Python notebook, one per chapter; nothing is typed in by hand. Simulations use fixed random seeds, so they repeat exactly. Where a chapter works something out by hand, it also shows that QuantLib gets the same answer.

The notebooks are free to download from github.com/RatesWithBip/curriculum, in the `options-from-zero` folder, and each has an *Open in Colab* button. They run as they are in Google Colab: each installs QuantLib and writes its own data files.

About the data

The share is Part 1's: it costs \$100, pays no dividends and has a volatility of 20%, and the interest rate is 3.60% a year. Our running example is still the one-year call struck at \$105, worth \$7.39. Chapters 10 and 11 reuse the illustrative AUD/USD, SOFR, AONIA and BBSW market built in the earlier *AUD Swaps & Curves* and *SOFR, FX & Cross-Currency Curves* books. The one exception is Chapter 9, which measures the realised volatility of the Australian dollar from the Federal Reserve Board's published exchange rates.

What comes next

Part 3 takes apart the assumption that each option has one volatility: the smile, the volatility surface, and the models built to explain it.

Free, and staying free

The videos, the notebooks and this book are free. If they help you, you can support the channel with a coffee at ko-fi.com/rateswithbip.

Options are risky. This book is for education only and is not financial advice.

Random Walks and Brownian Motion

Shrink a coin-flip walk's steps and it becomes Brownian motion. Its spread grows with the square root of time, and its squared moves add up to exactly the time elapsed.

From coin flips to paths

Part 1's tree moved the share up or down once per step. Take a walk that starts at zero and, at each of n steps in a year, moves up or down by $\sqrt{1/n}$ with equal chances. Each step adds $1/n$ to the variance, so after a year the variance is exactly 1, however many steps we take. With 4 steps a year the walk is Part 1's tree, one path at a time; with 16 and then 252 the path turns jagged. In the limit it is jagged everywhere and has no gaps: **Brownian motion**, written W_t .

The spread grows with the square root of time

20,000 simulated paths with daily steps: the average stays near zero (0.010 after a year), but the paths fan out.

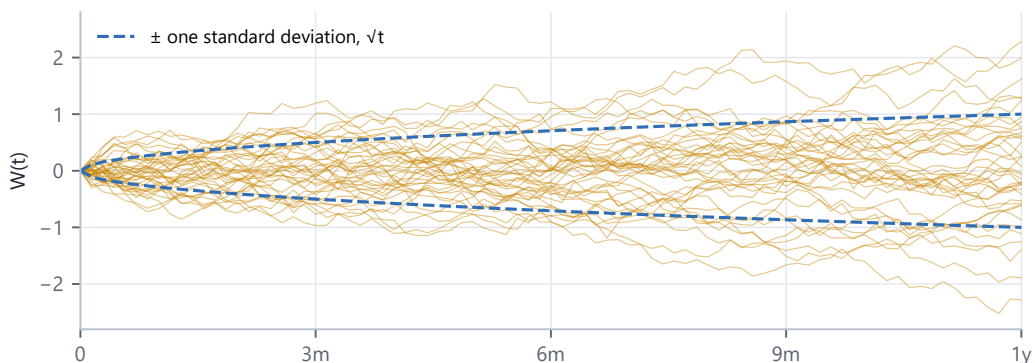


Figure 1.1. Forty of the simulated paths, with one standard deviation either side, $\pm\sqrt{t}$.

The variance grows in step with time, so the spread grows with its square root.

AFTER	VARIANCE	STANDARD DEVIATION	$\sqrt{t}$
3 months	0.250	0.500	0.500
6 months	0.509	0.713	0.707
1 year	1.010	1.005	1.000

Four times as long gives twice the spread. That is why Part 1 scaled a daily volatility to a yearly one with a square root. It also busts a common myth: randomness does not "average out". The average stays put, but the spread keeps growing, so the longer you wait, the less you know.

Infinitely wiggly

Add up the *size* of every move, ignoring direction. With n steps each move is about $\sqrt{1/n}$ in size, so the distance travelled grows like $n\sqrt{1/n} = \sqrt{n}$, without limit, while the endpoint stays in the same place.

Averages over 1,000 simulated paths.

STEPS IN A YEAR	DISTANCE TRAVELLED	$\sqrt{(2N/\pi)}$	DISTANCE FROM START AT THE END
12	2.7	2.8	0.80
52	5.8	5.8	0.84
252	12.7	12.7	0.82
10,000	79.8	79.8	0.79

Squared moves add up to time

Now add up the *squares* of the moves. Take one path with 10,000 steps and measure it with coarser and coarser steps. As the steps shrink, the sum of squared moves settles on 1, the time elapsed, and its wobble from path to path shrinks like $\sqrt{2/n}$:

Quadratic variation: the squared moves add up to the time elapsed.

STEPS	$\sum(\Delta W)^2$, ONE PATH	WOBBLE OVER 1,000 PATHS	$\sqrt{(2/N)}$
10	0.951	0.463	0.447
100	0.872	0.148	0.141
1,000	0.983	0.045	0.045
10,000	0.982	0.015	0.014

This is the **quadratic variation** of Brownian motion, $\sum (\Delta W)^2 \rightarrow t$, usually written $(dW)^2 = dt$. It looks like a curiosity. It is the fact that Itô's lemma, in Chapter 3, and much of Part 2 rests on.

A little history

Louis Bachelier modelled prices on the Paris exchange with random walks in his thesis *Théorie de la spéculation*, published in 1900. That was 5 years before Albert Einstein's 1905 paper explaining Brownian motion, the jiggling of particles suspended in a liquid. When it comes to the mathematics, finance got there first.

Bip's rule. Small random moves: the spread grows with $\sqrt{t}$, and squared moves add up to t .

Try this. If the variance after one year is 1, what is the standard deviation after 4 years?

Sources

Companion notebook: `s3e11_brownian_motion.ipynb` . Every number in this chapter is computed by it.

1. Numdam, *Théorie de la spéculation* (L. Bachelier), *Annales scientifiques de l'École Normale Supérieure*, 3e série, tome 17 (1900), 21-86. <https://www.numdam.org/articles/10.24033/asens.476/>
2. Wiley, *Über die von der molekularkinetischen Theorie der Wärme geforderte Bewegung von in ruhenden Flüssigkeiten suspendierten Teilchen* (A. Einstein), *Annalen der Physik* 17 (1905), 549-560. <https://doi.org/10.1002/andp.19053220806>

Slopes, Curvature and Taylor Series

A maths pit stop. Near a point, a curve is a straight line plus a bend. For an option, the line's slope is delta and the bend is gamma.

Last chapter's question

Variance grows with time, so after 4 years it is 4, and the standard deviation is its square root, 2.

The slope

Price our one-year \$105 call for a whole range of share prices. Today the share is at \$100 and the call is worth \$7.39. The straight line that just touches the curve there, the **tangent**, has slope 0.5144: for every dollar the share moves, the call moves about 51 cents. That slope is the first derivative, and for an option it is **delta**, Part 1's $N(d_1)$. The notebook measures it by nudging the share a cent either way:

$$\text{slope} \approx \frac{C(S+h) - C(S-h)}{2h}$$

The curvature

Further from \$100 the line misses, because the price curve bends upward. How fast the slope itself changes is the **curvature**, the second derivative:

$$\text{curvature} \approx \frac{C(S+h) - 2C(S) + C(S-h)}{h^2} = 0.0199$$

For an option this number is **gamma**. Adding the bend gives the **Taylor series** to second order:

$$C(S + \Delta S) \approx C(S) + \text{slope} \cdot \Delta S + \frac{1}{2} \text{curvature} \cdot (\Delta S)^2$$



Figure 2.1. The call's price, the tangent line, and the line plus the bend.

Because the curve bends upward, the call gains more when the share rises than it loses when the share falls by the same amount. That lopsidedness is what an option buyer pays for, and Chapter 5 puts a price on it.

How good is it?

Change in the call's price for share moves of \$1, \$5 and \$10 either way.

SHARE MOVES	TRUE CHANGE	LINE ONLY	LINE + BEND
-\$10	-\$4.13	-\$5.14	-\$4.15
-\$5	-\$2.32	-\$2.57	-\$2.32
-\$1	-\$0.50	-\$0.51	-\$0.50
+\$1	+\$0.52	+\$0.51	+\$0.52
+\$5	+\$2.82	+\$2.57	+\$2.82
+\$10	+\$6.09	+\$5.14	+\$6.14

For a \$1 move all three agree within a cent; for \$10 the line alone is well short and the bend brings it back within a few cents. What is left over shrinks like the *cube* of the move: from \$5 to \$10 the error grows about 9.2 times. So for small moves the slope is enough; for big ones, add the curvature.

In the notebook

For an option, the recipe reads: the change in price is about delta times the share's move, plus a half gamma times the move squared. Keep an eye on that move squared. In the next chapter it is where randomness sneaks in.

Bip's rule. Slope for small moves; add curvature for big ones.

Try this. The share falls \$10. Is the line-only estimate of the call's change too high or too low?

Companion notebook: `s3e12_taylor.ipynb` . Every number in this chapter is computed by it.

Itô's Lemma, Without the Pain

Because squared random moves add up to time, randomness plus curvature makes drift. For prices, the typical outcome grows at $\mu - \sigma^2/2$, not μ .

Last chapter's question

Too low. The line says the call loses \$5.14; it really loses only \$4.13. The bend, $\frac{1}{2}\Gamma(\Delta S)^2$, is always positive, so it always adds something back.

A puzzle

A share rises 50%: \$100 becomes \$150. Then it falls 50%: \$150 becomes \$75. The average of the two returns is zero, yet you have 75% of what you started with. In log returns, $\ln 1.5 + \ln 0.5 = -0.2877$. Ups and downs of the same size don't cancel.

Taylor meets randomness

For any smooth function $f(S)$, Chapter 2's Taylor series gives

$$df \approx f'(S) dS + \frac{1}{2} f''(S) (dS)^2$$

In ordinary calculus $(dS)^2$ is negligible. Here it isn't. With $dS = \mu S dt + \sigma S dW$ and Chapter 1's $(dW)^2 = dt$, the squared move is $(dS)^2 \approx \sigma^2 S^2 dt$: it adds up over time like the drift does. In the notebook, a year of daily returns squared adds up to 0.0400, the volatility squared, while the squared drift parts add up to only 0.00003. So

$$df = \left(f' \mu S + \frac{1}{2} f'' \sigma^2 S^2 \right) dt + f' \sigma S dW$$

This is **Itô's lemma**, after the Japanese mathematician Kiyosi Itô: randomness plus curvature makes drift.

The log price

Take $f(S) = \ln S$, so $f' = 1/S$ and $f'' = -1/S^2$:

$$d \ln S = \left(\mu - \frac{\sigma^2}{2} \right) dt + \sigma dW$$

The log price grows at $\mu - \sigma^2/2$, not μ . That $-\sigma^2/2$ is **volatility drag**: with an expected return of 8% and volatility of 20%, the drag is 2% and the log price grows at 6%. The notebook checks it without taking a single log along the way: it steps the share price forward with simple daily returns, then measures the average log growth: 6.0%.

Mean versus median

A million one-year outcomes with the same μ and σ :

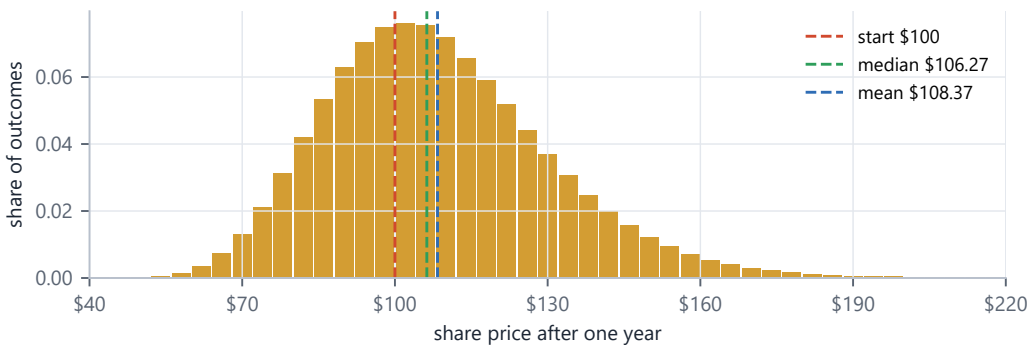


Figure 3.1. A million simulated one-year outcomes, with the start, the median and the mean.

The **mean** grows at the full μ : \$108.37. The **median**, the typical outcome, grows at $\mu - \sigma^2/2$: \$106.27. The outcomes are lopsided: a few big winners lift the mean, and 38% of the paths finish below where they started. Which busts a common myth: the expected return is not what a typical investor gets.

It's in Black-Scholes

The same term sits inside d_2 :

$$d_2 = \frac{\ln(S/K) + (r - \sigma^2/2) T}{\sigma \sqrt{T}}$$

Under the risk-neutral odds the share's median grows at $r - \sigma^2/2$, 1.6% a year for our trade, and $N(d_2)$ asks how likely the share is to finish above the strike.

Bip's rule. Volatility drags the typical outcome down by $\sigma^2/2$.

Try this. A 40% volatility share with an 8% expected return: what is its median growth rate?

Sources

Companion notebook: `s3e13_ito.ipynb`. Every number in this chapter is computed by it.

1. American Mathematical Society, *On Stochastic Differential Equations (K. Itô)*, *Memoirs of the American Mathematical Society No. 4 (1951)*. <https://bookstore.ams.org/memo-1-4/>

The Black-Scholes Equation from Hedging

Hold the option and sell delta shares. Over a short step the randomness cancels, so the book must earn the risk-free rate. That one condition is the Black-Scholes equation, and the expected return drops out.

Last chapter's question

The drag is half the variance, $0.40^2/2 = 8\%$, so the median grows at $8\% - 8\% = 0\%$. The typical outcome goes nowhere.

The hedged book

Own the option, worth $V(S, t)$, and sell Δ shares short: $\Pi = V - \Delta S$. Itô's lemma for V :

$$dV = \frac{\partial V}{\partial t} dt + \frac{\partial V}{\partial S} dS + \frac{1}{2} \frac{\partial^2 V}{\partial S^2} \sigma^2 S^2 dt$$

so $d\Pi = dV - \Delta dS$. Choose $\Delta = \partial V / \partial S$, the option's slope, and the dS terms, the only random part, cancel:

$$d\Pi = \left(\frac{\partial V}{\partial t} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} \right) dt$$

The proof in numbers

20,000 scenarios for tomorrow's share price, with an expected return of 8%. Unhedged, the call's value moves by a standard deviation of \$0.54. Hedged with delta shares (and a day's interest on the cash), it moves by \$0.015: the hedge removes 97% of the risk.

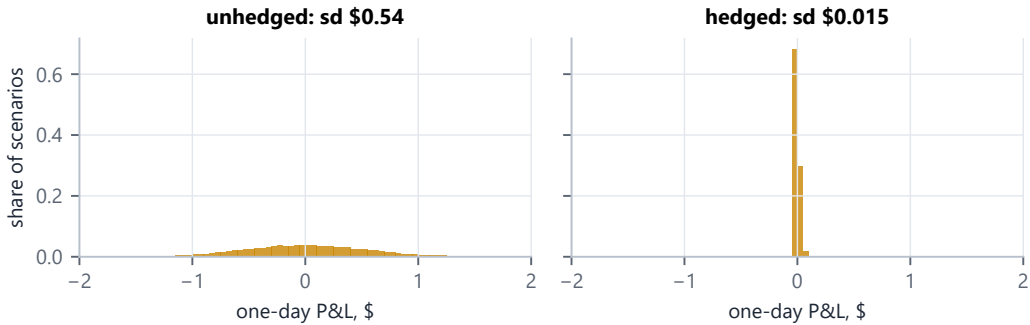


Figure 4.1. One day's profit or loss on the call, unhedged and hedged with delta shares.

No risk, so the risk-free return

A portfolio with no risk must earn the risk-free rate. If it earned more you'd borrow to buy it; if less, you'd sell it and lend. So $d\Pi = r\Pi dt = r(V - \Delta S) dt$, and setting the two expressions for $d\Pi$ equal gives the **Black-Scholes equation**:

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV = 0$$

It is a partial differential equation (PDE). Part 1's formula satisfies it: with the derivatives measured by nudging, the left-hand side is $-4.4\text{e-}08$.

Where did the expected return go?

It was in the share's moves, and it cancelled with them. That is the third time: arbitrage gave one fair price whatever the odds (Part 1, Chapter 3), risk-neutral odds priced without it (Part 1, Chapter 6), and now hedging removes it. The option's price does not depend on where the share is expected to go.

Solve it on a grid

The equation can be solved without the formula. Put the share price on a grid of points, start from the payoff at expiry, and step back to today, replacing each derivative with differences between neighbouring points: **finite differences**. The notebook's explicit scheme uses a grid from \$0 to \$250; QuantLib's `FdBlackScholesVanillaEngine` uses its own grid in log price.

Finite differences converge to the formula's price.

GRID POINTS	OUR TIME STEPS	OUR GRID	QUANTLIB	FORMULA
25	28	\$7.417	\$7.460	\$7.390
100	445	\$7.377	\$7.394	\$7.390
400	7,112	\$7.389	\$7.390	\$7.390

The explicit finite-difference solver in the notebook

```
def fd_explicit(points, s_max=250.0):
    '''Explicit finite differences for a call: step back from the payoff, one small time step at a time.'''
    ds = s_max / points
    grid = np.linspace(0.0, s_max, points + 1)
    dt_max = ds ** 2 / (vol ** 2 * s_max ** 2) # stability limit of the explicit scheme
    n_t = math.ceil(T / (0.9 * dt_max))
    dt = T / n_t
    v = np.maximum(grid - K, 0.0)
    for j in range(1, n_t + 1):
        tau = j * dt # time to expiry after this step
        i = slice(1, -1)
        gamma_ = (v[2:] - 2 * v[1:-1] + v[:-2]) / ds ** 2
        delta_ = (v[2:] - v[:-2]) / (2 * ds)
        v[i] = v[i] + dt * (0.5 * vol ** 2 * grid[i] ** 2 * gamma_ + r * grid[i] * delta_ - r * v[i])
        v[0], v[-1] = 0.0, s_max - K * math.exp(-r * tau)
    return float(np.interp(S, grid, v)), n_t
```

This is how banks price options that have no formula.

Bip's rule. Hedge away the randomness, and what's left must earn the risk-free rate.

Try this. In the one-day test, why isn't the hedged spread exactly zero?

Companion notebook: `s3e14_bs_pde.ipynb`. Every number in this chapter is computed by it.

Delta, Gamma, Theta

Delta is the hedge, gamma is how fast the hedge changes, and theta is the daily cost of owning gamma. For a hedged option, gamma is paid for with theta.

Last chapter's question

We set the hedge once and left it for a whole day; the argument needs it adjusted continuously. Re-run the test with the hedge held for an hour instead, and the risk left over is about 5 times smaller.

Three ways to get the Greeks

Delta $\partial V / \partial S$, **gamma** $\partial^2 V / \partial S^2$ and **theta** $\partial V / \partial t$ (calendar time: negative when the option loses value as time passes), by the formula, by nudging the inputs, and by QuantLib's analytic engine:

Greeks of the one-year \$105 call. Theta per year; the nudge moves time forward one day.

	FORMULA	NUDGING	QUANTLIB
delta	0.5144	0.5144	0.5144
gamma	0.0199	0.0199	0.0199
theta	-5.5726	-5.5751	-5.5726

The Greeks in options_part2.py

```
def bs_greeks(s, k, t, r, vol):
    """Price and Greeks of a European call on a share with no dividends.
    delta per $1 of share price, gamma per $1 squared, vega per 1.00 of volatility (100 vol
    points),
    theta per year (negative: value lost as time passes), rho per 1.00 of interest rate."""
    d1 = (math.log(s / k) + (r + vol ** 2 / 2) * t) / (vol * math.sqrt(t))
    d2 = d1 - vol * math.sqrt(t)
    disc = math.exp(-r * t)
    return {"price": s * norm_cdf(d1) - k * disc * norm_cdf(d2),
            "delta": norm_cdf(d1),
            "gamma": norm_pdf(d1) / (s * vol * math.sqrt(t)),
            "vega": s * norm_pdf(d1) * math.sqrt(t),
            "theta": -s * norm_pdf(d1) * vol / (2 * math.sqrt(t)) - r * k * disc *
            norm_cdf(d2),
            "rho": k * t * disc * norm_cdf(d2)}
```

Delta and gamma

With the share at \$100, a call struck at \$80 has delta 0.919, our \$105 call 0.514, and a call struck at \$130 only 0.151. With a month left, the curve turns into a cliff. Gamma, how fast delta changes, peaks near the strike and grows as expiry approaches: for a \$100-strike call it is 0.0192 with a year left, 0.0689 with a month and 0.1437 with a week, about 7 times the one-year figure.

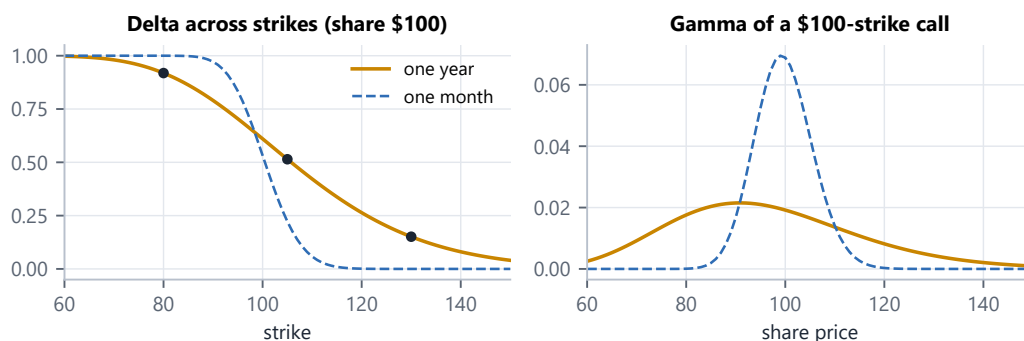


Figure 5.1. Delta across strikes and gamma across share prices, with a year and with a month to expiry.

Theta and the trade-off

Our call's theta is -5.57 a year, about 1.5 cents a day. Written with the Greeks, Chapter 4's equation is an identity:

$$\Theta + \frac{1}{2}\sigma^2 S^2 \Gamma + rS \Delta - rV = 0$$

For our call: $-5.57 + 3.99 + 1.85 - 0.27 = 0$ (to $1.1\text{e-}16$). The interest terms are small, so to a good approximation $\Theta \approx -\frac{1}{2}\sigma^2 S^2 \Gamma$: owning gamma, the curvature that gains from any move, costs theta every day.

The break-even move

Own the call and hedge it with delta shares for one day. Any big move makes money through gamma; time costs theta. If nothing moves, the book loses about 1 cent. The two balance at a move of $\sigma S \sqrt{1/365}$, about \$1.05, either way.

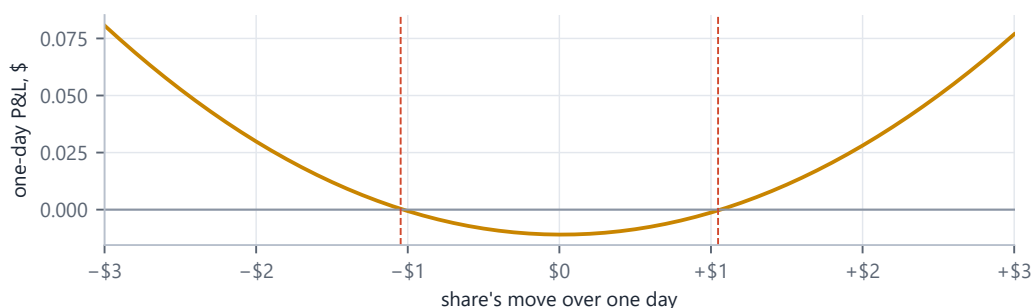


Figure 5.2. One day's P&L of the delta-hedged long call against the share's move, by exact repricing. The dashed lines mark $\pm\sigma S \sqrt{1/365}$.

Which busts a myth: theta is not free money for option sellers. A hedged seller collects theta on every quiet day but is short gamma, so every big move costs them. At a fair price, theta is exactly what the gamma risk is worth.

Bip's rule. Gamma is paid for with theta.

Try this. Who wants a quiet market: the option buyer or the seller?

Companion notebook: `s3e15_delta_gamma_theta.ipynb`. Every number in this chapter is computed by it.

Vega, Rho, and Explaining a Day's P&L

$P\&L \approx \Delta \cdot dS + \frac{1}{2} \Gamma \cdot dS^2 + \text{vega} \cdot d\sigma + \theta \cdot dt$. If the explained part doesn't match the actual change, something is missing from the risk.

Last chapter's question

The seller. A hedged seller is short gamma and collects theta, so a quiet day is a small win and a big move is a loss.

Vega

Vega is the change in price per unit of volatility, $S n(d_1) \sqrt{T}$. Traders quote it per vol point (one percentage point of volatility): \$0.40 for our call. Like gamma it is biggest near the strike, but it grows with time to expiry: a one-month call's vega is only \$0.09 per point.

Rho

Rho is the change in price per unit of interest rate, $Ke^{-rT} N(d_2)$: our call gains \$0.44 for a 1% rise in rates, because the strike is paid later and its present value falls. For a one-month call it is only \$0.02. Rho matters for long-dated options, and a great deal for options on interest rates themselves (Chapter 11).

The day

Overnight, the share rises, volatility rises half a point and a day passes.

	YESTERDAY	TODAY
Share price	\$100.00	\$101.50
Volatility	20.0%	20.5%
Time to expiry	365 days	364 days
Interest rate	3.60%	3.60%
Call price	\$7.390	\$8.369

Repriced, the call is up \$0.979. Why? Explain it with the Greeks, a second-order Taylor series in the share price and first-order in the rest:

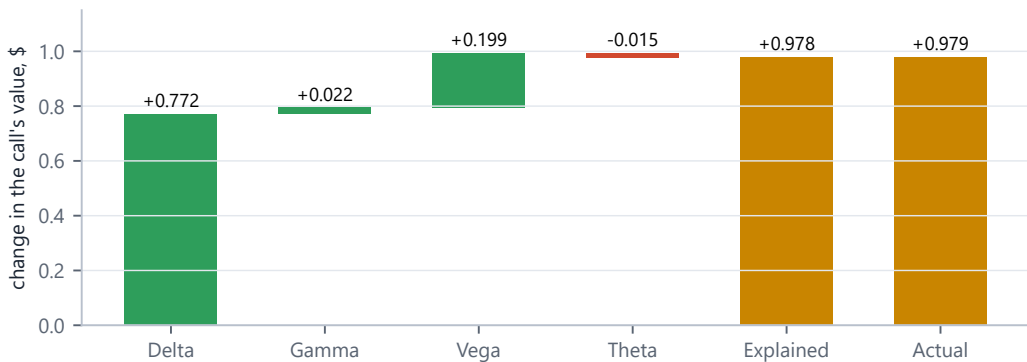


Figure 6.1. The day's change in the call, explained by delta, gamma, vega and theta, against the actual change.

Delta explains most of it, \$0.772; gamma adds \$0.022 for the size of the move; vega \$0.199 for the half point of volatility; theta takes 1.5 cents. The Greeks explain \$0.978 of the actual \$0.979: the **residual** is \$0.001.

Bigger days, bigger residuals

The terms the Greeks leave out, the share's move cubed and the cross terms between share and volatility moves, grow much faster than the moves themselves:

Today's moves scaled by 1/2, 1, 2 and 4; one day passes in each.

SHARE MOVE	VOL MOVE	ACTUAL	GREEKS	RESIDUAL
+\$0.75	+0.25 pts	+\$0.476	+\$0.476	\$0.000
+\$1.50	+0.50 pts	+\$0.979	+\$0.978	+\$0.001
+\$3.00	+1.00 pts	+\$2.019	+\$2.016	+\$0.003
+\$6.00	+2.00 pts	+\$4.220	+\$4.227	-\$0.007

On a violent day, or for options close to expiry where gamma is huge, the Greeks can miss by a lot, and desks reprice in full.

Why desks do this every day

P&L attribution is the daily proof that the risk numbers describe the book. A small residual means they do. A big residual means something is missing: a risk nobody measures, a wrong input, or a move too big for the Taylor series. It is the same check as explaining a swap book's P&L with its DV01 in *AUD Swaps & Curves*.

Bip's rule. If you can't explain the P&L, you don't know your risk.

Try this. The share is flat, volatility rises one point and a day passes. Roughly what is the P&L on our call?

Companion notebook: `s3e16_pnl_attribution.ipynb`. Every number in this chapter is computed by it.

Monte Carlo Pricing

Price = the average of discounted payoffs over simulated risk-neutral outcomes. The error bar shrinks like $1/\sqrt{N}$: a hundred times more paths buys ten times more accuracy.

Last chapter's question

Vega per point, \$0.40, minus a day of theta, \$0.02: roughly \$0.38.

The recipe

Part 1 showed that an option's price is its average payoff under the risk-neutral odds, discounted. So compute that average directly:

1. Draw the share price at expiry many times, growing at the risk-free rate: $S_T = S e^{(r-\sigma^2/2)T+\sigma\sqrt{T}Z}$, with Z a standard normal draw.
2. Pay $\max(S_T - K, 0)$ on each.
3. Discount, and average.

About 43% of the outcomes finish above the \$105 strike. This is **Monte Carlo**, named after the casino.

Monte Carlo in options_part2.py

```
def monte_carlo_call(s, k, t, r, vol, n, rng):
    """Price = the average discounted payoff over n risk-neutral outcomes, with its standard
    error."""
    z = rng.standard_normal(n)
    s_t = s * np.exp((r - vol ** 2 / 2) * t + vol * math.sqrt(t) * z)
    pv = math.exp(-r * t) * np.maximum(s_t - k, 0.0)
    return float(pv.mean()), float(pv.std(ddof=1) / math.sqrt(n))
```

Converging

Every estimate comes with a **standard error**: the standard deviation of the discounted payoffs divided by $\sqrt{N}$.

Monte Carlo estimates of the call; the formula gives \$7.39. One run per row, fixed seed.

PATHS	ESTIMATE	± STANDARD ERROR	MISS IN STANDARD ERRORS
1,000	\$7.537	0.419	0.35
10,000	\$7.397	0.127	0.05
100,000	\$7.415	0.040	0.62
1,000,000	\$7.409	0.013	1.53

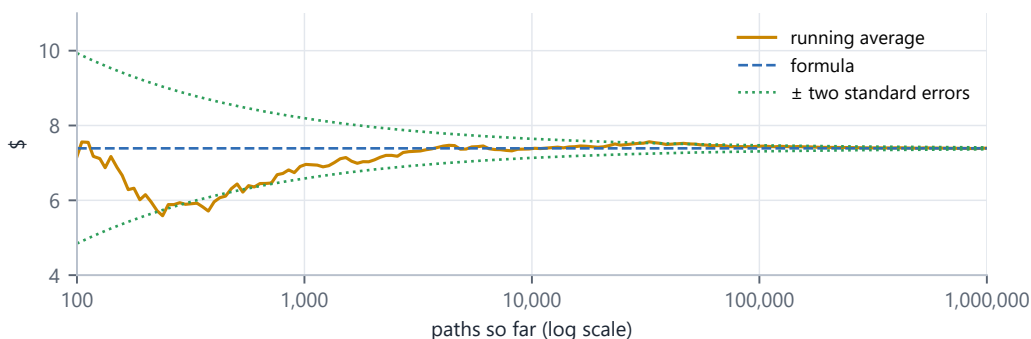


Figure 7.1. One long run: the running average of discounted payoffs, against the formula $\pm$ two standard errors.

The error bar

The standard error times $\sqrt{N}$ is constant, about \$12.89, the spread of a single discounted payoff. So the error shrinks like $1/\sqrt{N}$: a hundred times the paths buys ten times the accuracy, and pinning our call to within a cent would take about 1.7 million paths. Monte Carlo is easy to write but slow to converge.

QuantLib's `MCEuropeanEngine` does the same, with its own random numbers: \$7.36 $\pm$ 0.040, within its error bar of the formula.

A trick to speed it up

Antithetic variates: use each random draw twice, as Z and as $-Z$, and average the pair. When one path ends high its twin ends low, so their payoffs partly cancel and the average wobbles less. With the same 100,000 payoffs, the error bar falls from 0.040 to 0.033.

So why bother?

For our plain call the formula wins: exact and instant. Most options have no formula: an option that pays on the average price over a month, one on a basket of shares, one whose payoff depends on the route the price takes. For those you change step two, the payoff, and the recipe still works.

Bip's rule. Monte Carlo: easy to write, slow to converge, honest about its error.

Try this. An error bar of ± 0.4 with 1,000 paths: how many paths for ± 0.04 ?

Companion notebook: `s3e17_monte_carlo.ipynb`. Every number in this chapter is computed by it.

Hedging in Practice: Implied vs Realised Volatility

A delta-hedged option is a bet on realised versus implied volatility. Hedging more often shrinks the noise, not the expected profit.

Last chapter's question

Ten times the accuracy needs a hundred times the paths: 100,000.

The experiment

Sell the one-year \$105 call at 20% volatility and collect \$7.39. Hedge it: hold delta shares (computed at 20%), rebalanced every week or every day, borrowing and lending cash at the risk-free rate. Meanwhile the share moves with its *realised* volatility. Simulate 4,000 years and count the profit at expiry, in today's dollars. (The share grows at the risk-free rate here, to keep the story about volatility.)

The hedging simulator in options_part2.py

```
def hedge_short_call(s, k, t, r, vol_implied, vol_realised, steps, n_paths, rng, hedge=True):
    """Sell the call at vol_implied and, if hedge, rebalance to delta shares (computed at
    vol_implied) `steps` times.
    The share moves with vol_realised and grows at r. Cash earns r. Returns each path's P&L
    at expiry in today's dollars."""
    dt = t / steps
    price = s * np.ones(n_paths)
    held = call_delta(price, k, t, r, vol_implied) if hedge else np.zeros(n_paths)
    cash = black_scholes(s, k, t, r, vol_implied) - held * price  # premium received,
    shares bought
    for i in range(1, steps + 1):
        price = price * np.exp((r - vol_realised ** 2 / 2) * dt + vol_realised *
        math.sqrt(dt) * rng.standard_normal(n_paths))
        cash = cash * math.exp(r * dt)
        if hedge and i < steps:
            new = call_delta(price, k, t - i * dt, r, vol_implied)
            cash = cash - (new - held) * price  # buy (or sell) the
            difference
            held = new
        pnl = cash + held * price - np.maximum(price - k, 0.0)  # sell the shares, pay
        the option's payoff
    return pnl * math.exp(-r * t)
```

No hedge, weekly, daily

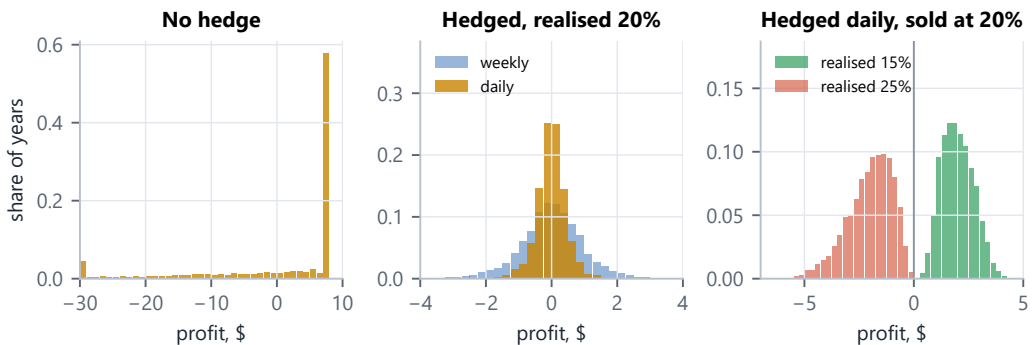


Figure 8.1. Profit at expiry of the sold call: unhedged; hedged weekly and daily when realised volatility is 20%; hedged daily when it is 15% or 25%.

Unhedged, the premium is profit in 70% of the years, but the worst 5% lose more than \$28 and the very worst \$88: a lottery ticket sold to someone else. Hedged weekly, the standard deviation falls to \$0.98; daily, to \$0.44, 2.2 times narrower, close to $\sqrt{252/52} = 2.2$. The average stays near zero (\$-0.01 daily): when the share's volatility is what we sold, the premium pays for the hedging, no more and no less. Hedging more often makes the result more certain, not bigger.

The twist

Same daily hedge, but the share realises 15% instead of 20%: on average we make \$1.99, and every simulated year makes money. At 25%, we lose \$2.01 on average. Same option, same hedge; only the share's actual volatility changed.

Why: theta earned, gamma paid

Each day the hedged seller collects theta, priced at the implied volatility, and pays for gamma according to how far the share actually moved (Chapter 5):

$$\text{daily P\&L} \approx \frac{1}{2} \Gamma S^2 \left(\sigma_{\text{implied}}^2 dt - \left(\frac{dS}{S} \right)^2 \right)$$

Summed over the year this is about $\text{vega} \times (\sigma_{\text{implied}} - \sigma_{\text{realised}})$: $39.9 \times 5\% = \$1.99$, right where the simulations landed. Summed day by day along each simulated path, the formula tracks the simulated profit with a correlation of 0.995.

What traders say

Selling volatility means selling options and hedging the delta: you win if the market moves less than the price implied. **Buying volatility** is the reverse. Once hedged, up or down no longer matters; only how much.

Bip's rule. Hedged options trade volatility, not direction.

Try this. You bought options at 20% volatility and hedged them. Realised volatility was 25%. Profit or loss?

Companion notebook: `s3e18_hedging_vol.ipynb`. Every number in this chapter is computed by it.

Implied Volatility: The Price Is Quoted as a Vol

Volatility is the one input you can't look up, so the market runs the formula backwards. Traders quote volatility, and the formula turns it into dollars.

Last chapter's question

A profit. The buyer is long gamma: wilder moves pay more than the theta costs, about $\text{vega} \times 5 \text{ points} = \1.99 on our call.

Price against volatility

Every Black-Scholes input can be looked up except one. The call's price rises steadily with volatility: \$3.40 at 10%, \$7.39 at 20%, \$11.37 at 30%. The curve only goes up, so each price corresponds to exactly one volatility, and we can run the formula backwards.

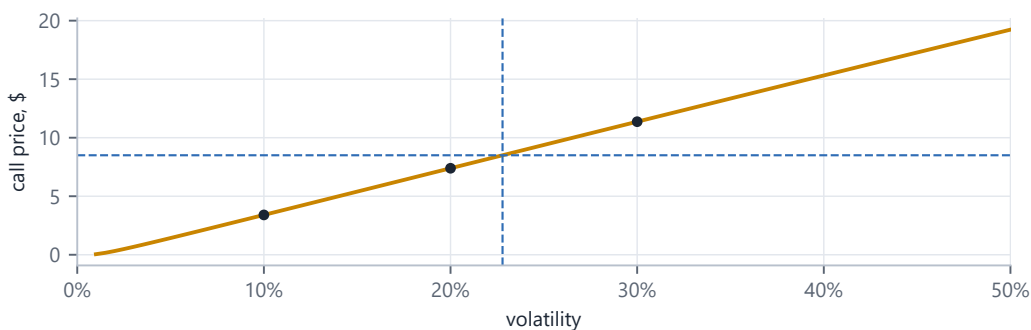


Figure 9.1. The call's price against volatility. The dashed lines read off the volatility for a price of \$8.50.

Run it backwards

The market quotes our call at \$8.50. **Bisection:** start with a range that surely contains the answer, try the middle, keep the half where the answer must be, repeat.

The first ten halvings for a price of \$8.50.

STEP	LOW %	HIGH %	TRY %	PRICE AT TRY	KEEP
1	0.01	300.00	150.01	\$54.39	lower half
2	0.01	150.01	75.01	\$28.79	lower half
3	0.01	75.01	37.51	\$14.34	lower half
4	0.01	37.51	18.76	\$6.89	upper half
5	18.76	37.51	28.13	\$10.63	lower half
6	18.76	28.13	23.45	\$8.76	lower half
7	18.76	23.45	21.10	\$7.83	upper half
8	21.10	23.45	22.27	\$8.30	upper half
9	22.27	23.45	22.86	\$8.53	lower half
10	22.27	22.86	22.57	\$8.41	upper half

After 42 halvings the **implied volatility** is 22.7866%; QuantLib's `impliedVolatility` agrees to 2.8e-13. A price of \$6.00 implies 16.52%. Newton's method, which uses vega as the slope, gets there in a few steps (see the notebook).

Implied volatility by bisection in `options_part2.py`

```
def implied_vol(price, s, k, t, r, lo=0.0001, hi=3.0, tol=1e-12):
    """Run Black-Scholes backwards by bisection: halve [lo, hi] until the price fits. Returns
    (vol, halvings)."""
    n = 0
    while hi - lo > tol:
        mid = (lo + hi) / 2
        if black_scholes(s, k, t, r, mid) < price:
            lo = mid
        else:
            hi = mid
        n += 1
    return (lo + hi) / 2, n
```

Why quote a volatility?

A dollar price goes stale the moment the share moves: at the same 20%, our call is \$7.39 with the share at \$100 and \$7.91 at \$101, but its volatility quote is unchanged. Volatility also compares options that dollars can't: a \$100-strike call costs \$9.72 and ours \$7.39, both at 20%. Vega is the exchange rate: \$0.40 per vol point for our call.

Implied versus realised

Implied volatility is a price: what the market charges today for the moves to come. **Realised** volatility is a measurement of what happened: the Australian dollar's, over the year of H.10 data to 18 September 2026, was 7.54%. Chapter 8 showed that the gap between the two is the hedger's profit or loss.

If Black-Scholes described the market perfectly, every strike would imply the same volatility. In real markets they don't, and that is where Part 3 begins.

Bip's rule. The market quotes volatility; the formula converts it into dollars.

Try this. Our call's quoted price rises by \$0.80. Roughly how many vol points is that?

Sources

Companion notebook: `s3e19_implied_volatility.ipynb`. Every number in this chapter is computed by it.

1. Board of Governors of the Federal Reserve System, *H.10 Foreign Exchange Rates, daily data download (Australian dollar, series RXI\$US_N.B.AL)*. <https://www.federalreserve.gov/releases/h10/>

FX Options: Garman-Kohlhagen

An FX option is Black-Scholes where the foreign interest rate plays the dividend. Two rates, one forward, same formula.

Last chapter's question

At \$0.40 per vol point, \$0.80 is about 2 vol points.

The trade

An **AUD call / USD put**: the right to buy AUD 1,000,000 at 0.7200 US dollars per Australian dollar. The market is the illustrative one from *SOFR, FX & Cross-Currency Curves*, valued on 25 September 2026 with spot at 0.7100. The spot date is 29 September 2026; the option expires on 29 December 2026 and the currencies are delivered on 31 December 2026, two business days later. The premium is paid in US dollars on the spot date. Volatility: an illustrative 9%.

Two rates

Garman and Kohlhagen (1983) showed that Black-Scholes still works for currencies, with one change: holding Australian dollars earns the Australian interest rate, just as a share earns its dividends, so the foreign rate plays the dividend:

$$C = S e^{-r_{AUD}T} N(d_1) - K e^{-r_{USD}T} N(d_2)$$

From spot to delivery, continuously compounded, the USD rate (SOFR curve) is 3.88% and the AUD rate (AUD under USD collateral) 4.13%.

The forward

The two rates meet in one number, the forward, by covered interest parity:

$$F = S \times \frac{P_{AUD}}{P_{USD}} = 0.70956$$

4.4 points below spot, because the Australian rate is higher. Written on the forward, Garman-Kohlhagen is Black's formula (Chapter 11), with the volatility over today to expiry and discounting from spot to delivery:

$$C = P_{USD} [F N(d_1) - K N(d_2)], \quad d_{1,2} = \frac{\ln(F/K) \pm \frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}}$$

Which busts a myth: the forward is not the market's forecast of the exchange rate. It is spot adjusted by the two interest rates, and the option prices off it for the same reason Part 1's share grew at the risk-free rate.

The price

The call costs USD 8,444. FX desks also quote it in **pips**, ten-thousandths of a US dollar per Australian dollar (84.4 pips), or as a percentage of the AUD amount (1.19%). The put with the same strike costs USD 18,783: the strike is above the forward.

The hedge

The **spot delta** is $P_{AUD}N(d_1) = 0.3799$: sell about AUD 379,899 in the spot market against the option. For AUD/USD the market's conventions are a premium paid in US dollars, *regular* (not premium-adjusted) delta, and spot delta out to one year (Reiswich and Wystup 2009, citing Clark). The forward delta, for a hedge with a forward, is $N(d_1) = 0.3839$.

Parity again, and QuantLib

Put-call parity, FX style: call – put = $P_{USD}(F - K)$. Both sides are –USD 10,339, to 5.6e-11. QuantLib has no separate Garman-Kohlhagen process in its Python build, so the notebook uses the Black-Scholes-Merton process with the AUD curve as the dividend curve. QuantLib discounts from today to expiry rather than spot to delivery; in its own convention it matches the hand formula to 1e-12, and the two conventions differ by USD 7 on AUD 1 million.

Bip's rule. FX options: two rates, one forward, same formula.

Try this. If Australian interest rates rose, would this AUD call get cheaper or dearer?

Sources

Companion notebook: `s3e20_fx_options.ipynb`. Every number in this chapter is computed by it.

1. Elsevier, *Foreign currency option values* (M. B. Garman and S. W. Kohlhagen), *Journal of International Money and Finance* 2(3), 231-237 (1983). [https://doi.org/10.1016/S0261-5606\(83\)80001-1](https://doi.org/10.1016/S0261-5606(83)80001-1)
2. Frankfurt School of Finance & Management (EconStor), *FX Volatility Smile Construction* (D. Reisch and U. Wystup), *CPQF Working Paper No. 20* (2009). <https://www.econstor.eu/bitstream/10419/40186/1/613825101.pdf>

Rates Options: Black-76 and Bachelier

A caplet is a call on a future BBSW fixing. Price it on the forward with Black-76, or with normal volatility in basis points with Bachelier, which still works when rates go negative.

Last chapter's question

Cheaper. A higher Australian rate lowers the forward $F = S P_{AUD}/P_{USD}$, and the call prices off the forward.

The caplet

A **caplet** pays $\max(L - K, 0) \times \tau \times N$, where L is 3-month BBSW set at the start of a future period, τ the period's year fraction (ACT/365F) and N the notional. It is set in advance and paid in arrears. A **cap** is a string of caplets: a borrower paying BBSW buys one to put a ceiling on their rate. Ours: AUD 10 million, strike 4.00%, on the period from 27 September 2027 to 29 December 2027 ($\tau = 0.2548$).

The forward

From the Series 1 curves: the forward 3-month BBSW rate for that period, from the BBSW 3M curve, is 4.2488%, and the payment date's discount factor on the AONIA curve is 0.9513: forecast on BBSW, discount on AONIA, the dual-curve rule.

Black-76

Fischer Black (1976) priced options on forwards and futures by treating the forward as lognormal. For a caplet the forward rate plays the share:

$$\text{caplet} = N \tau P [F N(d_1) - K N(d_2)], \quad d_{1,2} = \frac{\ln(F/K) \pm \frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}}$$

with T the time to the fixing and a *lognormal* volatility, a percentage of the rate. At an illustrative 18%, the caplet is worth AUD 10,602.

Bachelier: volatility in basis points

Bachelier (1900, Chapter 1) let prices move by normal amounts. For rates that means volatility in basis points:

$$\text{caplet} = N \tau P \left[(F - K) N(d) + \sigma_N \sqrt{T} n(d) \right], \quad d = \frac{F - K}{\sigma_N \sqrt{T}}$$

The normal volatility that gives Black's price is 74.1bp; the rule of thumb $\sigma_N \approx \sigma F$ gives 76.5bp. QuantLib's `blackFormula`, `bachelierBlackFormula`, `BlackCapFloorEngine` and `BachelierCapFloorEngine` all agree with the hand formulas to a fraction of a cent.

Black's and Bachelier's formulas in `options_part2.py`

```
def black76(forward, strike, t, vol, df, call=True):
    """Black's formula on a forward: df x [F N(d1) - K N(d2)] for a call. FX options (Garman-Kohlhagen) and caplets."""
    sd = vol * math.sqrt(t)
    d1 = (math.log(forward / strike) + sd ** 2 / 2) / sd
    d2 = d1 - sd
    if call:
        return df * (forward * norm_cdf(d1) - strike * norm_cdf(d2))
    return df * (strike * norm_cdf(-d2) - forward * norm_cdf(-d1))

def bachelier(forward, strike, t, normal_vol, df, call=True):
    """Bachelier's formula: the forward moves by normal_vol x sqrt(t) in absolute terms (it can go below zero)."""
    sd = normal_vol * math.sqrt(t)
    d = (forward - strike) / sd
    if call:
        return df * ((forward - strike) * norm_cdf(d) + sd * norm_pdf(d))
    return df * ((strike - forward) * norm_cdf(-d) + sd * norm_pdf(d))
```

Same price here, different wings

The two models agree at the 4.00% strike by construction. Away from it they don't:

Caplets on AUD 10 million: Black at 18%, Bachelier at 74.1bp.

STRIKE %	BLACK	BACHELIER	BLACK - BACHELIER
3.00	AUD 30,431	AUD 30,617	-186
3.50	AUD 19,366	AUD 19,631	-264
4.00	AUD 10,602	AUD 10,602	0
4.50	AUD 4,962	AUD 4,549	+413
5.00	AUD 2,013	AUD 1,470	+543

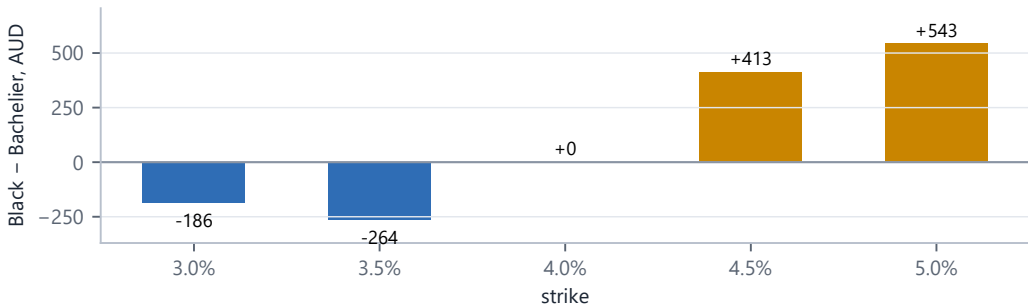


Figure 11.1. Black minus Bachelier, by strike.

Black's lognormal forward has a long upper tail, so high strikes are dearer, and it can't go below zero, so low strikes are cheaper. Lognormal and normal volatility are not two ways of writing the same thing: they are different models, each implying its own pattern of volatility across strikes. Choosing a model is choosing a smile, where Part 3 begins.

Negative rates

Policy rates have gone below zero: the ECB's deposit facility rate was negative from 2014 to 2022, reaching -0.50%, and the Bank of Japan applied -0.1% from 2016 until 2024. Black's formula needs $\ln(F/K)$, which doesn't exist for a negative forward. Bachelier's doesn't care: a 0% caplet on a forward of -0.20% at 60bp normal volatility is worth AUD 3,814 on AUD 10 million.

Part 2 in one picture

The maths of randomness (Brownian motion, Taylor, Itô) led to the Black-Scholes equation by hedging, with the expected return dropping out again. The Greeks measure the risk and explain the P&L; grids and Monte Carlo price what formulas can't; a hedged

option is a bet on volatility, which the market quotes directly. And the same machinery prices options on the Australian dollar and on BBSW.

Bip's rule. Rates options: Black on the forward, or Bachelier in basis points.

Try this. The Part 2 checkpoint in the appendix.

Sources

Companion notebook: `s3e21_caps_black_bachelier.ipynb`. Every number in this chapter is computed by it.

1. Numdam, *Théorie de la spéculation (L. Bachelier)*, *Annales scientifiques de l'École Normale Supérieure, 3e série, tome 17 (1900)*, 21-86. <https://www.numdam.org/articles/10.24033/asens.476/>
2. Elsevier, *The pricing of commodity contracts (F. Black)*, *Journal of Financial Economics* 3(1-2), 167-179 (1976). [https://doi.org/10.1016/0304-405X\(76\)90024-6](https://doi.org/10.1016/0304-405X(76)90024-6)
3. Bank of Japan, *Introduction of "Quantitative and Qualitative Monetary Easing with a Negative Interest Rate" (29 January 2016)*. https://www.boj.or.jp/en/mopo/mpmdeci/mpr_2016/k160129a.pdf
4. Bank of Japan, *Changes in the Monetary Policy Framework (19 March 2024)*. https://www.boj.or.jp/en/mopo/mpmdeci/mpr_2024/k240319a.pdf
5. European Central Bank, *Key ECB interest rates (history of the deposit facility rate)*. https://www.ecb.europa.eu/stats/policy_and_exchange_rates/key_ecb_interest_rates/html/index.en.html

Checkpoint and glossary

Part 2 checkpoint

Try these before Part 3. The answers follow.

1. Explain today's P&L on an option with delta 0.5, gamma 0.02, vega 0.4 per vol point and theta -0.015 a day: the share rose \$2 and volatility fell 1 point.
2. You sold an option at 20% volatility, hedged it daily, and realised volatility came in at 18%. Profit or loss, and roughly how much, with vega 40 (per 1.00 of volatility)?
3. Why does the AUD/USD call of Chapter 10 use the AUD interest rate at all, if the premium is paid in US dollars?

Answers.

1. Delta $0.5 \times 2 = 1.00$, gamma $\frac{1}{2} \times 0.02 \times 2^2 = 0.04$, vega $0.4 \times (-1) = -0.40$, theta -0.015 : in total +\$0.625.
2. A profit: the seller is short gamma, and the share moved less than the 20% priced in. About vega $\times$ the gap, $40 \times (0.20 - 0.18) = 0.80$.
3. Holding Australian dollars earns the Australian interest rate, like a dividend. The two rates set the forward, $F = S P_{AUD}/P_{USD}$, and the option prices off the forward.

Glossary

Antithetic variates. A Monte Carlo trick: use each random draw twice, as Z and $-Z$, so the pair's errors partly cancel.

Bachelier model. Prices (or rates) move by normal, absolute amounts; volatility is quoted in basis points for rates. Works for negative rates.

Black-76. Black-Scholes applied to a forward or futures price, with lognormal volatility; the standard formula for caplets and FX options written on the forward.

Black-Scholes equation. $\partial V / \partial t + \frac{1}{2} \sigma^2 S^2 \partial^2 V / \partial S^2 + rS \partial V / \partial S - rV = 0$: the condition that a delta-hedged option earns the risk-free rate.

Brownian motion. The limit of a random walk with ever-smaller steps: continuous, nowhere smooth, with variance equal to elapsed time.

Cap, caplet. A caplet pays $\max(L - K, 0) \times \tau \times N$ on one future rate fixing; a cap is a string of caplets.

Delta. The change in an option's price per \$1 move in the underlying; the number of shares that hedges it.

Finite differences. Solving a PDE on a grid, replacing derivatives with differences between neighbouring points.

Gamma. The change in delta per \$1 move; the option's curvature.

Garman-Kohlhagen. Black-Scholes for currency options, with the foreign interest rate as the dividend.

Implied volatility. The volatility that makes the Black-Scholes price equal the market price.

Itô's lemma. Taylor's series with $(dW)^2 = dt$: for a function of a random process, curvature times variance becomes drift.

Monte Carlo. Pricing by averaging discounted payoffs over simulated risk-neutral outcomes; its error shrinks like $1/\sqrt{N}$.

P&L attribution. Explaining a change in value with the Greeks; the **residual** is what they leave unexplained.

Pip. 0.0001 of the quoted currency per unit of the base currency: for AUD/USD, a ten-thousandth of a US dollar per Australian dollar.

Quadratic variation. The sum of squared moves of a path; for Brownian motion it equals the time elapsed.

Realised volatility. Volatility measured from past price moves.

Rho. The change in price per unit of interest rate.

Spot delta. For an FX option, the amount of foreign currency to trade in the spot market to hedge it.

Standard error. The standard deviation of an estimate: for Monte Carlo, the payoffs' standard deviation divided by $\sqrt{N}$.

Theta. The change in price as time passes, everything else fixed.

Vega. The change in price per unit of volatility, often quoted per vol point.

Volatility drag. The $-\sigma^2/2$ by which the median (typical) outcome grows more slowly than the mean.

Further reading

These books cover the same ground from different angles; none of this book's examples come from them.

- John C. Hull, *Options, Futures, and Other Derivatives*: the Greeks, Monte Carlo, FX and interest-rate options.
- Sheldon Natenberg, *Option Volatility and Pricing*: hedging, the Greeks and volatility trading from a trader's point of view.
- Steven E. Shreve, *Stochastic Calculus for Finance II: Continuous-Time Models*: the rigorous version of Chapters 1 to 4.
- Paul Wilmott, *Paul Wilmott Introduces Quantitative Finance*: the PDE approach and finite differences.
- Iain J. Clark, *Foreign Exchange Option Pricing: A Practitioner's Guide*: FX option conventions in depth.